PROGYA PAROMITA BISWAS

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EDUCATION

Vellore Institute of Technology 2023 – 2027
B.Tech in Computer Science and Engineering; CGPA: 8.5/10
Dr. Graham's Homes 2010 – 2023
Secondary (ICSE, 2020–2021) | Higher Secondary (ISC, 2022–2023)

RESEARCH EXPERIENCE

Comparative Analysis of Transfer Learning Models with Explainable AI [LINK] May 2025 – October 2026
DOI: 10.5281/zenodo.23835020

- Evaluated six pretrained architectures (ResNet-50, VGG16, DenseNet-121, ConvNeXt-Tiny, ViT-B/16, and DINOv2) across frozen, partially fine-tuned, and fully fine-tuned transfer learning strategies for 13-class nail disease classification using 1,008 images and grouped stratified 5-fold cross-validation.
- Achieved 88.1% accuracy and 0.857 Macro-F1 with fully fine-tuned DINOv2; applied Grad-CAM and Attention Rollout for model interpretability and potential classification bias analysis.

Energy-Aware Parallel Computation of SHAP Explanations [LINK] Aug 2026 - Present

- Benchmarked Serial, Multiprocessing, Threading, Joblib, and Ray strategies for TreeSHAP and KernelSHAP, completing 972 runs across 324 configurations with CPU time and energy measured using CodeCarbon.
- Analyzed workload-dependent time-energy trade-offs across Random Forest, XGBoost, SVM-RBF, and Logistic Regression to generate time-optimal, energy-optimal, and balanced CPU scheduling recommendations.

Reinforcement Learning-Based Multi-Robot Navigation [LINK] Feb 2026 – Apr 2026
Research Paper, Robotics Lab, VIT Vellore

- Implemented autonomous mobile robot navigation using DQN and PPO reinforcement learning.
- Integrated a Waveshare robot chassis, Jetson Nano, RPLiDAR, and camera modules with ROS2/Gazebo for autonomous navigation; designed the sensor fusion and real-time communication architecture.

TECHNICAL PROJECTS

Real-Time Distance Estimation using Monocular Vision [LINK] Jan 2026 – Present

- Developed a system that estimates object distance in real time from a single RGB camera, implementing monocular depth and distance estimation algorithms.
- Optimized the inference pipeline for real-time performance; paper submitted to IEEE ICAITPR 2026 and currently under review. Potential applications include autonomous systems and assistive technology for visually impaired individuals.

HorizonEye — Vision-Based Horizon Detection & Height Estimation Aug 2026 – October 2026

- Developed a camera-based flight perception system using sky-ground segmentation, Otsu thresholding, and horizon line fitting to estimate aircraft roll and horizon dip angle.
- Implemented camera and spherical-Earth geometry to estimate flight altitude using Python, OpenCV, and NumPy.

Sound Energy Harvesting System using Piezo Sensors Jan 2026 – Feb 2026

- Designed a piezoelectric system to convert ambient sound vibrations into usable electrical energy for low-power applications.
- Implemented energy storage and power optimization circuits to improve energy utilization and system efficiency.

LEADERSHIP

Team Lead, Sound Energy Harvesting System using Piezo Sensors Jan 2026 – Feb 2026

- Conceived the idea and led a 2-member team to develop a piezoelectric energy harvesting system that converts ambient sound vibrations into usable electrical energy for low-power applications.

Team Lead, Reinforcement Learning-Based Robot Navigation, *Robotics Lab, VIT Vellore* [LINK] Feb 2026 – Apr 2026

- Led a 3-member research team, under faculty guidance, to develop an RL-based multi-robot navigation system.

Team Lead, CookBook Web Application [LINK] Feb 2026 – Apr 2025

- Led a 4-member team to design and build a responsive recipe-sharing web application in ReactJS, with interactive features for recipe sharing and community engagement, integrating frontend components with backend services.

TECHNICAL SKILLS

Languages: C++, C, HTML, CSS, Python, Java, MySQL, Oracle Database
Intelligence: Machine Learning, Deep Learning, Computer Vision, Image Processing, TensorFlow, Keras
Robotics: ROS2, Gazebo, Sensor Integration, Real-time Systems
Tools: Git, GitHub, Linux (Ubuntu), VS Code, MATLAB, Jupyter Notebook, Google Colab, Cisco Packet Tracer, Keil, Arduino IDE, ModelSim, StarUML